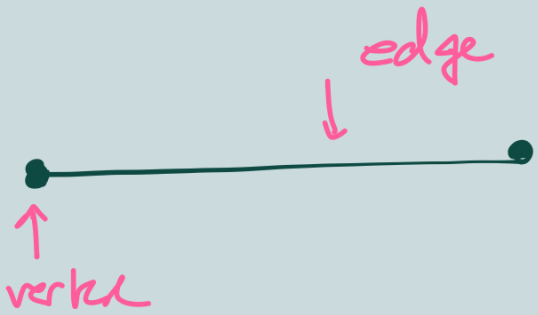
Graph



Trivial graph | only vertex, no edge

Null graph | n vertices, no edge.

Directed graph | edge এর direction আছে

Self-loop |

Simple graph | self-loop & multiedge না থাকলে
(দুইয়ের বেশি edge same end point থাকলে না)

Multigraph | self loop নাই, multiedge আছে

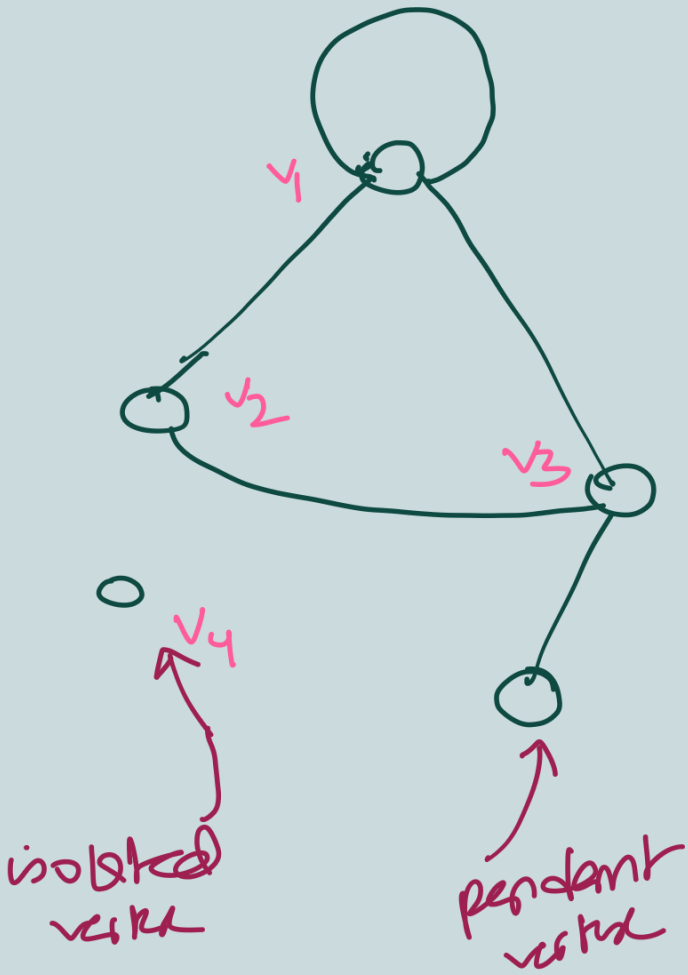
Pseudograph | self-loop + multiedge

Incidency and Adjacency

v_1, v_2 adjacent to e_0

e_k is incident of v_1, v_2

Degree of vertex → $d(v)$



self-loop ⇒ degree → 2

সরাসরি connected edge → ১ degree

$d(v_1) = 2 + 2$

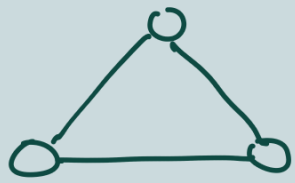
$d(v_2) = 2$

$d(v_4) = 0$

$d(v_5) = 1$

Complete graph

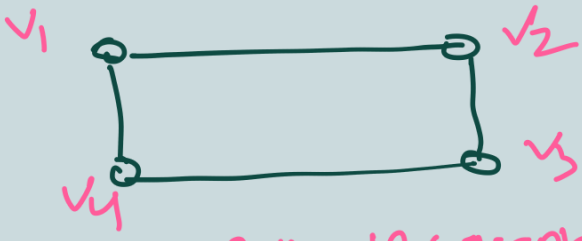
each vertex is connected to every other vertex.



Regular graph

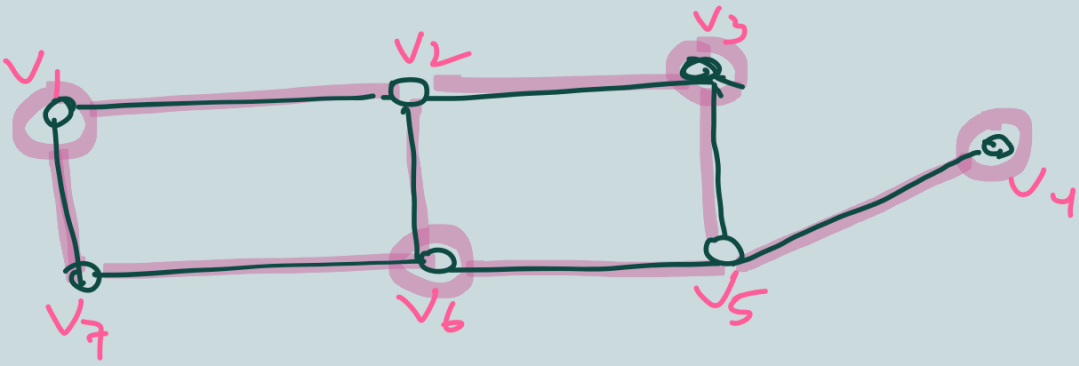
every vertex has same degree.

$d(v) = k$; k-regular graph.



2 regular graph

Bigraph/ Bipartite



$X = \{v_1, v_6, v_3, v_4\}$

$Y = \{v_2, v_7, v_5\}$

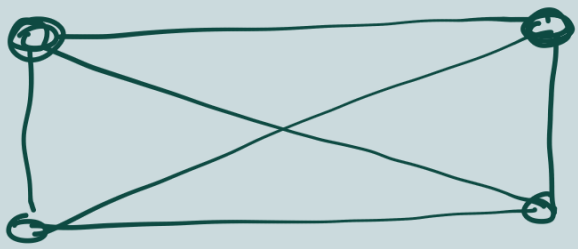
graph কে partition করে 2 টি non-empty disjoint set বানানো হয় edge এর one end → X and another end → Y.

Connected graph

প্রত্যেক 2 vertices connected থাকলে

+

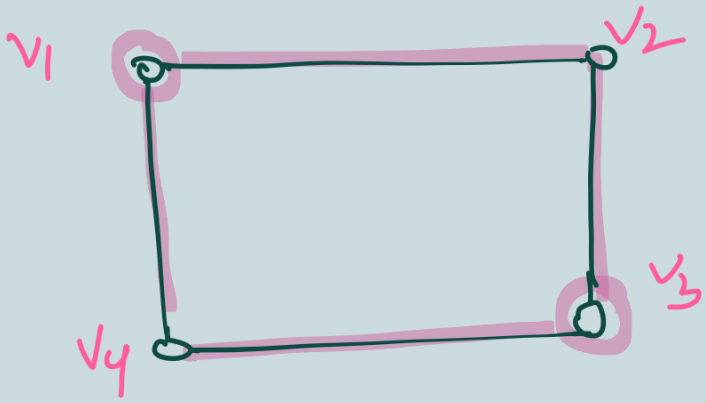
undirected graph



$\boxed{\text{graph connected} \rightarrow \text{not bipartite}}$

Complete bipartite graph

every element of X is connected to all element of Y.



$X = \{v_1, v_3\}$

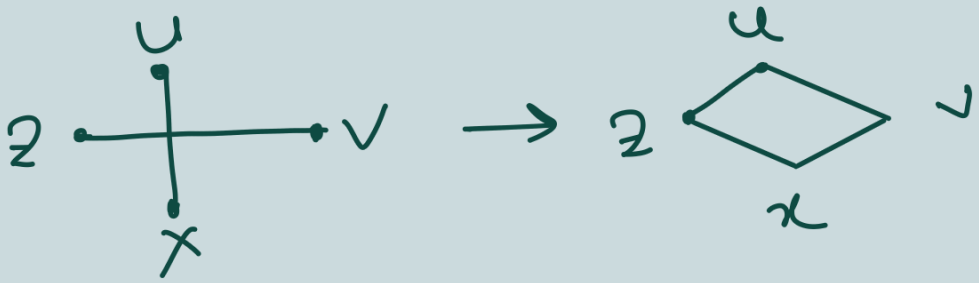
$Y = \{v_2, v_4\}$

Subgraph

একটা graph এর subset. তার edges.

Complement of graph

vertices এর মধ্যে edge গুলো remove করে new edge বানানো

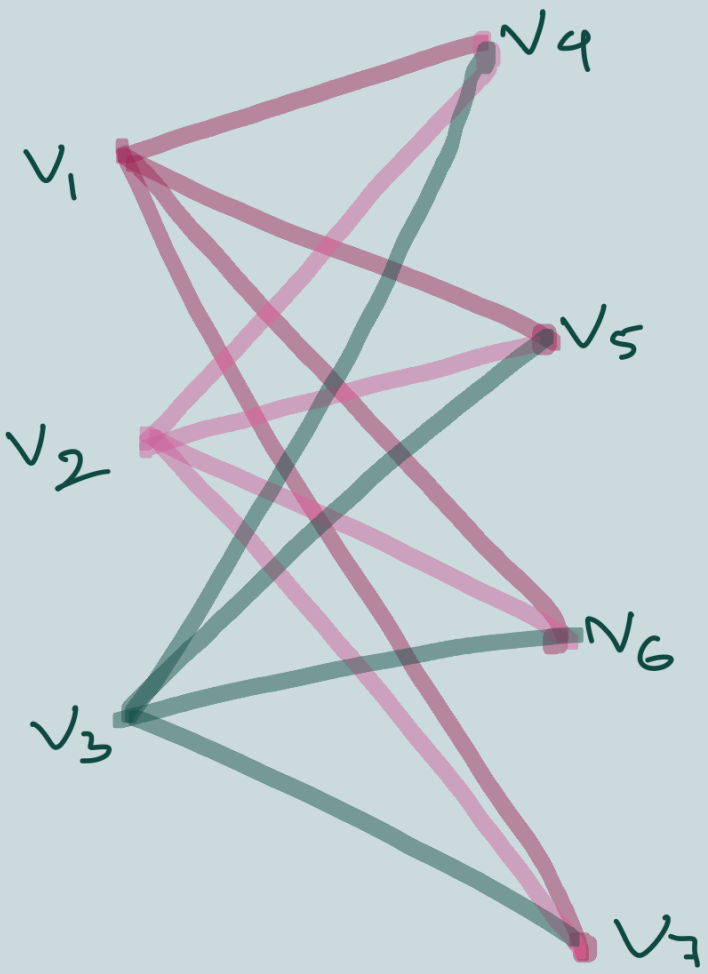


another set

* Draw K_{34} bipartite graph

one set

complete bipartite graph

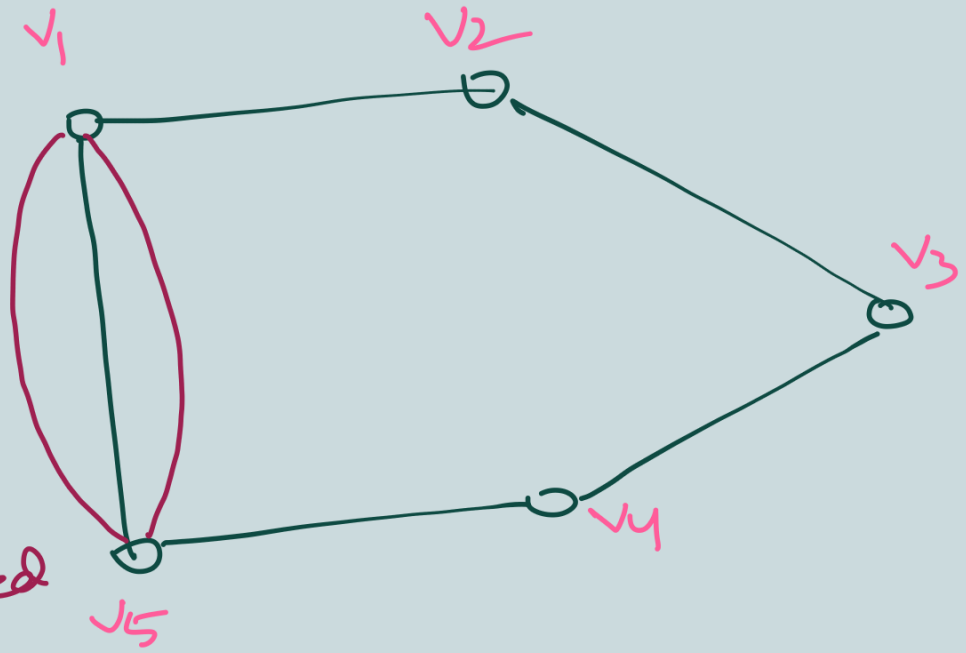


Handshaking theorem

sum of degree of vertices = 2 × number of edges

$$\sum_{i=1}^{n} \deg(v) = 2e$$

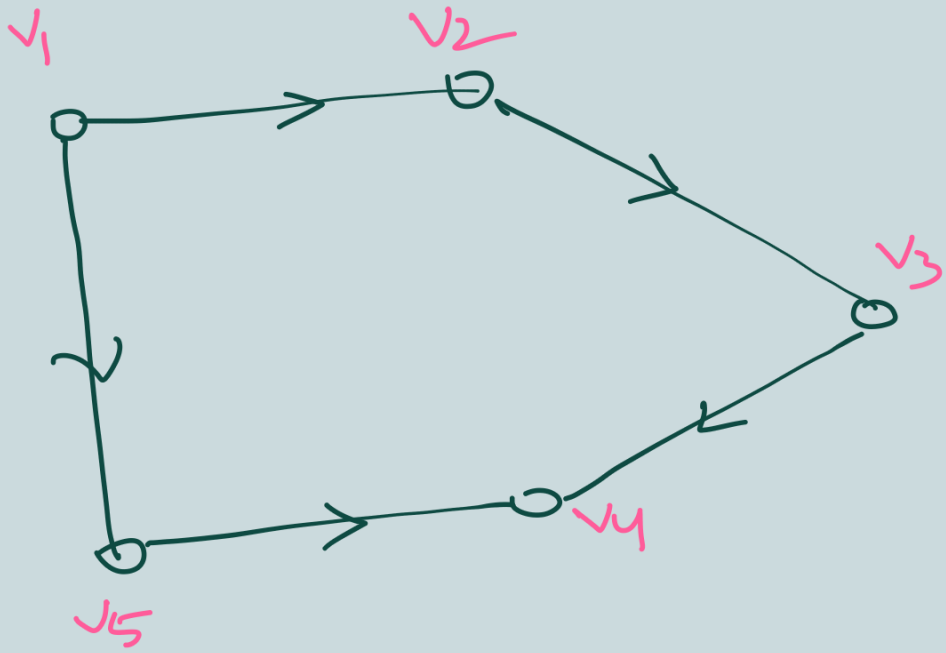
Matrix representation of graph



if self connected $= 1$

$$
\begin{array}{c|ccccc}
 & v_1 & v_2 & v_3 & v_4 & v_5 \\
\hline
v_1 & 0 & 1 & 0 & 0 & 3 \\
v_2 & 1 & 0 & 1 & 0 & 0 \\
v_3 & 0 & 1 & 0 & 1 & 0 \\
v_4 & 0 & 0 & 1 & 0 & 1 \\
v_5 & 3 & 0 & 0 & 1 & 0
\end{array}
$$

undirected graph হলে দুই side count করবে ; symmetric থাকবে



$$
\begin{array}{c|ccccc}
 & v_1 & v_2 & v_3 & v_4 & v_5 \\
\hline
v_1 & 0 & 1 & 0 & 0 & 1 \\
v_2 & 0 & 0 & 1 & 0 & 0 \\
v_3 & 0 & 0 & 0 & 1 & 0 \\
v_4 & 0 & 0 & 0 & 0 & 0 \\
v_5 & 0 & 0 & 0 & 1 & 0
\end{array}
$$